CODESCHOOL · TEACHER TRAINING PLAN · DRAFT

Teacher training plan

How we prepare and support teachers to teach Computational Thinking and AI in Classes 3-8, from the first orientation to a training platform

Prepared by Anoop Dobhal, CodeSchool (Bullseye Knowledge System Limited)

11 November 2026

Draft version 0.1

Builds on the April 2027 launch plan, the teacher books and the assessment books

Contents

About this plan

This plan sets out how CodeSchool trains and supports teachers in the 2027-28 session and beyond. It extends the 2-hour orientation in the launch plan into a year-long programme. Dates are targets, and costs are estimates to be checked. A separate online training platform will come later; section 10 lists what it needs to do so the work here carries straight into it.

1. Why teachers come first . . . 3
Principles . . . 3
2. Who we are training . . . 3
3. Teacher competency framework . . . 4
4. The training model . . . 4
Why not one long training before the session? . . . 5
5. Session designs . . . 5
5.1 Orientation (2 hours, all teachers of a school or cluster) . . . 5
5.2 Foundation workshop (1 day, by band: Classes 3-5 or 6-8) . . . 5
5.3 Teach-ahead sessions (90 minutes, before each block of chapters) . . . 5
5.4 Practice circles (90 minutes, monthly, about six a year) . . . 5
5.5 Classroom visits . . . 6
5.6 Leaders' briefing (45 minutes, principals and coordinators) . . . 6
6. Materials . . . 6
7. Year-1 calendar . . . 7
8. Capacity, scaling and certification . . . 7
Capacity check . . . 7
Growing champion teachers . . . 7
Certificates . . . 8
9. Measuring whether training works . . . 8
10. Requirements for the future training platform . . . 8
11. Team and budget (estimates) . . . 9
12. Main risks . . . 9
13. Next steps . . . 10

1. Why teachers come first

CodeSchool is a **teacher-led** programme. In the Offline Programme there are no student devices: the teacher, the workbook, the classroom kit and the teacher book do all the work. If a teacher is unsure, the CT&AI period is the first one to be skipped or turned into a maths period. So the aim of training is not just to explain the books. It is to make every teacher **confident enough to teach the next lesson tomorrow**, and to keep them confident all year.

Most teachers who will teach CT&AI were never taught it themselves. In Classes 3-5 it will usually be the class teacher or the maths teacher; in Classes 6-8, the maths teacher or the computer teacher. Many will worry about three things: "I am not a coder", "AI is too new for me", and "When will I find the time?". The plan answers each one:

Teacher's worry	What we do about it
"I am not a coder"	CT is taught unplugged, through maths chapters they already teach. Training lets them *do* each activity as a student first.
"AI is too new for me"	Classes 6-8 AI chapters are taught with cards and paper, not software. A short AI-literacy module covers just what they need, with plain-language answers to common student questions.
"When will I find the time?"	Every period is already scripted in the teacher book. Training is short, in school hours where possible, and counts towards the 50 hours of yearly professional development that NEP 2020 expects of teachers.
"How do I assess this?"	The assessment books give a one-page test per chapter with marking guides; training shows how to turn scores and observations into journal levels.

Principles

- **Do, then discuss.** Teachers experience each activity as students before they teach it.
- **Just in time.** Train for the next unit, not the whole year at once.
- **Close to the classroom.** Cluster sessions near schools, in school hours where possible, with follow-up in their own classrooms.
- **Peer-led over time.** Grow champion teachers who support colleagues, so support does not depend on a few trainers.
- **Measured.** We track confidence, classroom practice and student results, and change the training when they show a gap.

2. Who we are training

Group	Typical role	Starting point	What they most need
Primary teachers (Classes 3-5)	Class teacher or maths teacher	Strong at activity-based teaching; little CT background	The CT vocabulary (algorithm, loop, condition, variable, debugging), managing kit activities, the observation journal
Middle-school maths teachers (Classes 6-8)	Maths teacher	Strong maths; little CT or AI	Seeing the CT idea inside each maths chapter, the AI chapters, running 20-hour projects
Computer teachers (Classes 6-8)	ICT or computer science teacher	Knows software and some coding	Teaching *unplugged* and through maths; the AI chapters as concepts rather than tools; project assessment
Academic coordinators and heads of department	Plan timetables, observe lessons	Varies	The programme map, timetabling the hours, observing and supporting CT&AI lessons
Principals	Approve adoption and timetable	Varies	Why CT&AI matters, CBSE's expectations, what support the school will get, how progress is reported

Group	Typical role	Starting point	What they most need
Champion teachers (from year 1)	Experienced CodeSchool teachers	Secure in the programme	Coaching colleagues, running school-level sharing sessions

3. Teacher competency framework

Training is built around five competency areas. Each has three levels. A teacher should reach **Practitioner** in the areas for their classes during the first year.

Area	Starter (after orientation)	Practitioner (end of year 1)	Mentor (champion)
A. CT concepts	Explains algorithm, pattern, decomposition, loop and condition with an everyday example	Traces and debugs the programs in the workbook; links each CT idea to its maths chapter	Explains misconceptions and designs new examples
B. AI literacy and ethics (Classes 6-8)	Tells AI from automation; explains learning from examples	Teaches the four AI chapters for their class; handles bias, privacy and safety questions	Leads discussion of new AI news and tools with colleagues
C. Unplugged teaching	Runs one kit activity using the teacher-book script	Runs every period type smoothly: warm-up, guided activity, pair work, exit check; manages groups and materials	Adapts activities for large classes and mixed abilities
D. Assessment	Uses an exit check and records one observation	Uses the chapter tests, marking guides and journal levels; writes term summaries	Moderates marking across sections and coaches others
E. Projects (Classes 6-8)	Knows the project steps and rubric	Runs both 20-hour projects, including group work and presentations	Helps other teachers plan and assess projects

Area A applies to all teachers; B and E to Classes 6-8; C and D to all.

4. The training model

Training runs in four layers across the year, rather than as one event.

Layer	What	When	Length	Format
1. Orientation	The programme, the books and kit, and the first unit, taught hands-on	22 March-15 April 2027, before the first CT&AI period	2 hours (minimum, in the launch plan)	Cluster or in-school session
2. Foundation workshop	Competencies A-E in depth for each band, with practice teaching	April-May 2027, or as a second session in a school's in-service days	1 day (6 hours)	Cluster workshop, by class band
3. In-term support	Teach-ahead sessions before each block of units; monthly practice circles; classroom visits; a help line	All year	About 1-2 hours a month	Cluster, online call or school visit
4. Growth and certification	Certificate levels, champion teachers, an annual teacher meet	From term 2	Ongoing	Portfolio and observation

Hours per teacher. Orientation (2 h) + foundation workshop (6 h) + four teach-ahead sessions (4 x 1.5 h) + six practice circles (6 x 1.5 h) adds up to about **23 hours** in the first year: almost half the 50-hour yearly expectation, all of it used directly in class.

Why not one long training before the session?

Long one-off trainings are forgotten by the time a topic comes up in class. A short orientation gets teachers started on the first unit; the foundation day and teach-ahead sessions then arrive just before teachers need them.

5. Session designs

5.1 Orientation (2 hours, all teachers of a school or cluster)

Time	Activity	Outcome
0:00-0:15	Welcome; "What is computational thinking?" warm-up: teachers write the steps to make tea, then swap and follow each other's steps exactly	Feel what an algorithm and a bug are
0:15-0:35	The programme map: CBSE's CT&AI curriculum, hours for each class, how the chapters follow NCERT maths	Know what they will teach and when
0:35-1:15	**Model lesson**: the trainer teaches period 1 of chapter 1 for each band; teachers take part as students, using the workbook and kit	Seen and done one full period
1:15-1:35	Tour of the teacher book: the five-period plan, scripts, common mistakes, exit check, answer key; the assessment book	Can find everything for any lesson
1:35-1:50	Classroom routines: handing out and collecting kit pieces, pair roles, noise signal, what to do with fast finishers	Ready to manage the class
1:50-2:00	Plan the first week; questions; join the support group (optional, with consent)	A dated plan for the first CT&AI period

5.2 Foundation workshop (1 day, by band: Classes 3-5 or 6-8)

Session	Classes 3-5	Classes 6-8
1. CT ideas in depth (75 min)	Sequences, loops, conditions, variables and debugging through the term-1 chapters	Functions, lists, searching and sorting, number bases, through the term-1 chapters
2. AI literacy (60 min)	Optional short session: what AI is, using the Class 5 data chapters	Supervised, unsupervised and reinforcement learning; data, bias and ethics, using the four AI chapters
3. Practice teaching (90 min)	In small groups, each teacher teaches one 10-minute part of a lesson; peers give feedback with the observation checklist	Same, plus one AI activity
4. Assessment (45 min)	Marking the chapter tests together; agreeing journal levels	Same, plus the project rubrics
5. Projects (45 min)	Not needed	Planning the two 20-hour projects: groups, timeline, data sheets, presentations
6. Planning (30 min)	Term-1 timetable and the first teach-ahead date	Same

5.3 Teach-ahead sessions (90 minutes, before each block of chapters)

Four times a year, at the start of each half-term: teachers work through the next 3-4 chapters as students, look at the common mistakes in the teacher book, and share how the last block went. For Classes 6-8, one session is about the AI chapters and one is about each project.

5.4 Practice circles (90 minutes, monthly, about six a year)

Small groups of 8-15 teachers from nearby schools, led by a trainer or a champion teacher, using the same structure every time:

1. **Share** (20 min): one success and one problem from the last month.
2. **Look at student work** (30 min): marked tests or project work; agree on what "Secure" looks like.
3. **Solve** (25 min): pick one common problem, such as a large class or a weak reader, and plan a fix.
4. **Commit** (15 min): each teacher picks one thing to try before the next circle.

Circles can be held online where travel is hard.

5.5 Classroom visits

A trainer or champion visits each new teacher at least **once a term** to watch a CT&AI period, using a short observation checklist (clear objective, students active, kit used well, exit check done, journal kept). The feedback has two strengths, one next step and a follow-up date. Visits are supportive, not inspections, and the notes go to the teacher, not the school management, unless the teacher agrees.

5.6 Leaders' briefing (45 minutes, principals and coordinators)

What CT&AI is and why it matters; CBSE's expectations; timetabling (one or two periods a week for Classes 3-5; the 100-hour structure for Classes 6-8); what support teachers get; how to observe a lesson; how progress is reported to parents.

6. Materials

Material	Status	Use
Teacher books, Classes 3-8	**Ready** (drafts)	Scripts, plans, answers, journal pages
Assessment books, Classes 3-8	**Ready** (drafts)	Chapter tests, marking guides, record sheets
Specimen booklets	**Ready**	Leaders' briefing and school visits
Trainer guide (session plans, timings, materials lists, FAQs)	To write, February 2027	So any trainer or champion runs sessions the same way
Orientation and workshop slides, by band	To make, February-March 2027	Sessions in 5.1-5.3
Trainer demo kit (large-size cards and boards)	To make with the kit prototype	Model lessons in front of a group
Quick-reference cards (one per class: the year at a glance, CT words, classroom routines)	To make, March 2027	Kept in the teacher book
Model-lesson videos (5-8 minutes each, one per chapter type)	Record in pilot classes, January-March 2027, with written consent from the school and parents	Orientation, revision and later the platform
Misconception bank (common student mistakes and how to fix them)	Start from the teacher books; grow from pilots and circles	Teach-ahead sessions and the platform
AI-literacy explainer for teachers (8-10 pages)	To write, March 2027	Foundation workshop, Classes 6-8
Observation checklist and teacher self-assessment	To write, February 2027	Visits and measuring progress

7. Year-1 calendar

When	What	Who
Nov-Dec 2026	Agree this plan; include training in proposals and pricing; draft the competency framework into the trainer guide	Anoop
4 Jan-15 Mar 2027	**Pilot**: train the 2-3 pilot teachers with the orientation and workshop drafts; record model lessons; collect their feedback	Anoop, pilot teachers
Feb 2027	Hire one trainer (as in the launch plan); write the trainer guide, observation checklist and slides	Anoop, trainer
Mar 2027	Train-the-trainer rehearsal; book cluster venues; leaders' briefings in adopting schools	Trainer
22 Mar-15 Apr 2027	**Orientations** for every adopting teacher	Trainer, Anoop
Apr-May 2027	Foundation workshops by band; first practice circle	Trainer
Jul 2027	Teach-ahead 2; first classroom visits	Trainer
Aug-Sep 2027	Practice circles; visits continue; term-1 review of student results	Trainer
Oct 2027	Teach-ahead 3 (term 2); choose the first champion teachers	Trainer
Nov 2027-Jan 2028	Practice circles; second round of visits; champion training (2 days)	Trainer, champions
Feb 2028	Teach-ahead 4; certificates for teachers who complete the year	Trainer
Mar 2028	Annual teacher meet; year-1 report; plan year 2 with the training platform	Anoop, trainer

Exact dates will follow each region's school calendar and exam periods.

8. Capacity, scaling and certification

Capacity check

Assumption	Number
Adopting schools (the range used in the launch plan)	10-100
CT&AI teachers per school (about one per class, 3-8, shared across sections)	4-6
Teachers to train	40-600
Teachers per orientation	20-30
Orientations needed	2-30

One trainer can run two orientations a day in the 3.5-week window, so **one trainer covers up to about 30 orientations**. Beyond about 50 schools, a second trainer or a few experienced freelance trainers will be needed for the March-April peak, and practice circles should move to champion teachers.

Growing champion teachers

From term 2, pick one champion in each school or cluster: a teacher who is Practitioner level in every area for their band and wants to help others. Champions get a 2-day training (coaching, running circles, observing lessons), a certificate and recognition in their school, and they lead circles and new-teacher support in year 2. This is what lets the programme grow without a large trainer team.

Certificates

Certificate	Requirement
CodeSchool CT&AI Teacher: Starter	Orientation attended
Practitioner	Foundation workshop, at least four circles or teach-ahead sessions, one positive classroom observation, a short portfolio (one marked test with journal levels; for Classes 6-8, one project rubric)
Mentor (champion)	Practitioner, plus champion training and one term leading a circle

Certificates show the hours of training, so schools can count them towards teachers' yearly professional-development hours. If CBSE or a state body later recognises outside training providers, we will apply.

9. Measuring whether training works

Level	Measure	How	Target, year 1
Reach	Teachers oriented before their first CT&AI period	Attendance	100% of adopting teachers
Reaction	Session rating	Two-minute form at the end	Average 4 out of 5 or higher
Confidence	"I can teach next week's CT&AI lesson"	Same short survey before orientation, after the workshop and at year end	Most teachers rate themselves confident by the end of term 1
Practice	Lessons that meet the observation checklist	Classroom visits	Most observed lessons meet at least 4 of the 5 points
Student results	Average chapter-test scores and journal levels	Class record sheets (shared without student names)	Average at or above 6 out of 10
Continuation	Teachers and schools continuing in year 2	Renewals	High renewal; any drop explained

Teachers' personal data (names, phone numbers, survey answers) is collected only with consent, stored privately, never in this public repository, and handled in line with the Digital Personal Data Protection Act. Student results are used only as class averages.

10. Requirements for the future training platform

The platform will come later. Building the training this way now means its content moves straight across. The platform should:

- **Follow the competency framework** (section 3): learning paths for each band (Classes 3-5, 6-8) and for computer teachers, coordinators and champions.
- **Teach chapter by chapter**: for each chapter, a short video model lesson, the key idea in two minutes, common mistakes and a five-question self-check, released just before the chapter is due.
- **Work on a phone with low bandwidth**: short videos, downloadable PDFs, offline viewing; English and Hindi first, other languages later.
- **Support practice and feedback**: teachers upload a short clip or photo of student work and get feedback from a trainer or champion.
- **Run live sessions**: schedule practice circles and teach-ahead sessions, take attendance and send reminders, with teachers' consent.
- **Issue certificates and track hours**: automatic hour logs and certificates linked to the levels in section 8.
- **Give a trainer dashboard**: who has completed what, confidence scores, observation results, and which schools need a visit.
- **Link to the Online Programme**: one sign-in with the online teacher dashboard when schools take the Online Programme.

- **Protect privacy**: collect the minimum data, ask for consent, let teachers see and delete their data, and keep student data out of the training platform.

Phase	When	What
0. Now	2027-28 session	PDFs, model-lesson videos (shared privately with teachers), forms for feedback, video calls for circles
1. First version	Ready for 2028-29 orientations	Learning paths, chapter videos, self-checks, certificates, hour logs
2. Full platform	During 2028-29	Practice uploads with feedback, live-session scheduling, trainer dashboard, link to the Online Programme

11. Team and budget (estimates)

Item	Year-1 estimate	Notes
Lead trainer (from February 2027)	About Rs 5-7 lakh (est.)	Already in the launch plan
Extra trainers for the March-April peak (freelance)	About Rs 0.5-2 lakh (est.)	Only above about 50 schools
Travel and venues for cluster sessions	About Rs 1-3 lakh (est.)	Many sessions can be hosted by schools
Trainer demo kits, quick-reference cards, printing	About Rs 0.5-1 lakh (est.)	Alongside the kit print run
Model-lesson videos (phone, tripod, microphone, editing)	About Rs 0.5-1.5 lakh (est.)	Recorded in pilot classes
Champion recognition (certificates, annual meet)	About Rs 0.5-1 lakh (est.)	—
Training platform	Not in year 1	To be costed separately

Training is included in the price of the programme for adopting schools, which makes it a selling point in proposals. Extra in-school workshops could be offered as a paid option later.

12. Main risks

Risk	Early warning	Response
Schools cannot release teachers for training	Few sign-ups for workshops	Run sessions inside school in-service days or after school; offer the foundation day as two half-days or online
Teacher turnover in schools	New teachers mid-year	Short catch-up orientation, online or by a champion; recorded model lessons
One trainer is not enough	More than about 50 schools adopt	Contract freelance trainers in February; move circles to champions earlier
Training feels like extra work	Low attendance at circles	Keep sessions short and practical; count hours towards professional development; recognise good teaching publicly
Teachers are not confident with AI	AI questions dominate sessions	AI explainer, extra AI teach-ahead, and a question bank with simple answers
Visits feel like inspections	Teachers avoid them	Keep notes with the teacher; make visits supportive and agreed in advance

13. Next steps

By	Step
Fri 20 Nov 2026	Agree the training model (section 4) and include training in the school proposals
Fri 18 Dec 2026	Draft the orientation session and the observation checklist; plan the pilot-teacher training
Mon 4 Jan 2027	Train the pilot teachers before the pilot begins
Mon 1 Feb 2027	Lead trainer hired; trainer guide drafting starts
Mon 15 Mar 2027	Trainer guide, slides, quick-reference cards and demo kit ready; orientation dates booked
Thu 15 Apr 2027	Every adopting teacher oriented

© 2026 Bullseye Knowledge System Limited. All rights reserved.